

Big Data Tools and Ecosystem

Categories of Big Data tooling

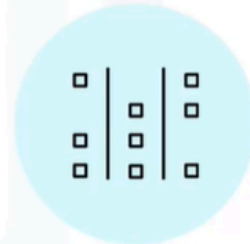


- Data Technologies Analytics and Visualization Business Intelligence Cloud service providers NoSQL Databases; and Programming Tools Let's take a look at each of these components in detail.

Data technologies

Data technologies allow enterprises to:

- Capture, process and share data at any scale and in any format
- Work with structured and unstructured data
- Leverage high-performance, parallel Big Data processing



Key technologies include Hadoop, HDFS, Spark, MapReduce, Cloudera, and Databricks

Analytics and Visualization

- Big Data analytics examines large amounts of data
- Analyzed data is visualized using graphs, charts, and maps



Popular analytics tools available are Tableau, Palantir, SAS, Pentaho, and Teradata

Business Intelligence

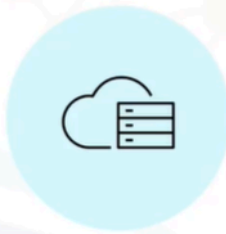
- Business intelligence (BI) offers a range of tools that provide a quick and easy way to transform data into actionable insights
- Such insights inform an organization's strategic and tactical business decisions

Examples include Cognos, Oracle, PowerBI, Business Objects, and Hyperion



Cloud providers

Offer fundamental infrastructure and support with shared resources including computing, storage, networking, and analytical software



NoSQL databases

NoSQL databases are best suited for Big Data processing:

- Store and process vast amounts of data at scale
- Store information in JSON documents instead of relational tables
- NoSQL database types include pure document databases, key-value stores, wide-column databases, and graph databases

Examples include MongoDB, CouchDB, Cassandra, Redis



Programming tools

Big Data programming tools:

- Perform large-scale analytical tasks and operationalize Big Data
- Provide all necessary functions for the Big Data life cycle

R, Python, SQL, Scala, and Julia are common programming tools

